Exp. no. 7
11/3/26

# FREQUENCY RESPONSE OF SERIES AND PARALLEL RESONANCE CIRCUITS

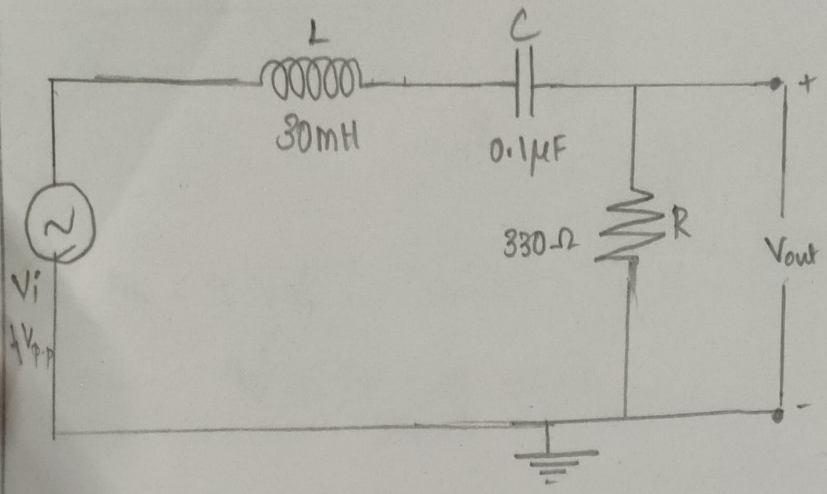
AIM :

To study the behaviour of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using Multisim simulator.

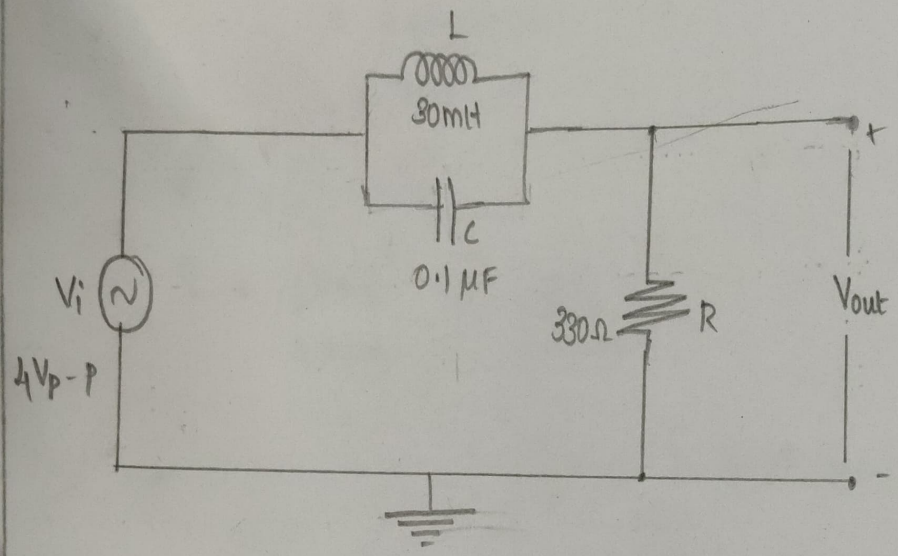
APPARATUS REQUIRED :

| S. NO | COMPONENTS | SPECIFICATIONS | QUANTITY |
|---|---|---|---|
| 1 | CRO | 10MHz | 1 |
| 2 | DIB | - | 1 |
| 3 | capacitors | 0.1 μF | 1 |
| 4 | Resistors | 330 Ω | 1 |
| 5 | Bread board | - | - |
| 6 | connecting wires | - | - |
| 7 | PC with Multisim simulator | - | - |

Circuit diagram :

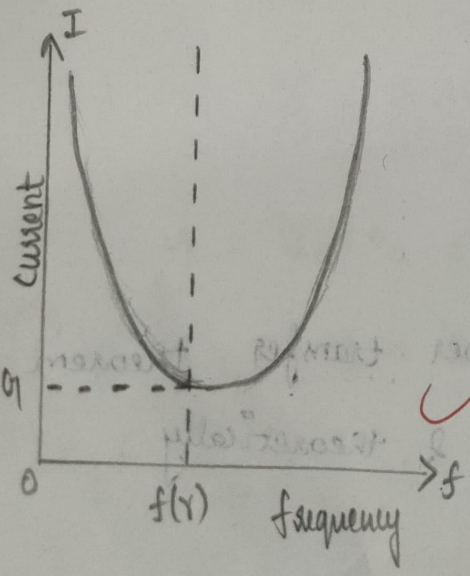



Series RLC circuit

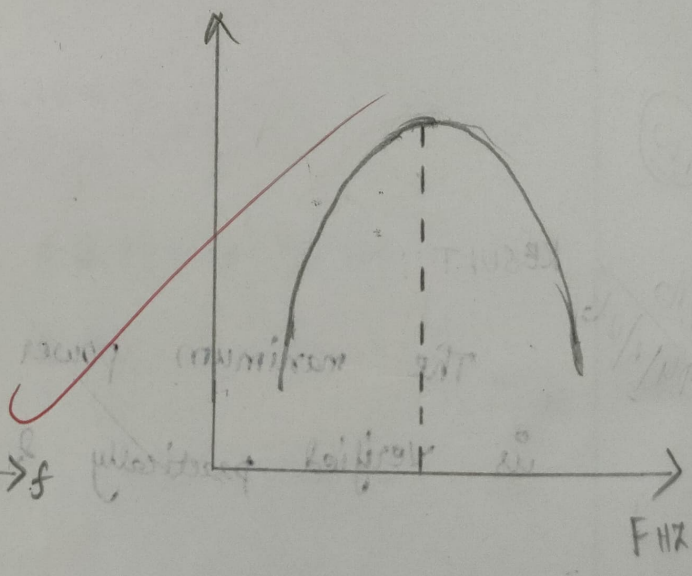



Parallel RLC circuit

Parallel Resonance




Series Resonance



TABULATION :

$V_{in} = 4\ V_{p-p}$

| SERIES RESONANCE | | | PARALLEL RESONANCE | | |
|---|---|---|---|---|---|
| Frequency (Hz) | output voltage (V) | output current (mA) | Frequency (Hz) | output voltage (V) | output current (mA) |
| 500 | 1.2 | 2 | 500 | 1.5 | 9 |
| 700 | 2.0 | 4 | 700 | 2.8 | 7 |
| 900 | 3.5 | 7 | 900 | 4.0 | 5 |
| (Resonance) 1000 | 5.0 | 10 | (Resonance) 1000 | 5.0 | 2 (minimum) |
| 1100 | 3.6 | 7 | 1100 | 4.1 | 5 |

Also o/p voltage

CALCULATION :

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{(30\times10^{-3})(0.1\times10^{-6})}} = \frac{1}{2\pi\sqrt{3\times10^{-9}}}$$

$$= \frac{1}{2\pi(5.47\times10^{-5})}$$

$$= 1000\ Hz$$

FORMULA :

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$BW = f_2 - f_1$$

$$Q = f_0 / BW$$

PROCEDURE :

i) Construct circuit shown in Figure 1 on breadboard

ii) Connect CRO ch 1 to input and ch 2 to output

iii) Set the input voltage to 4 Vp-p

iv) Vary the frequency from 500 Hz to 3 kHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage $V_m$ is the resonance frequency.

v) If $V_{cutoff} = \frac{V_m}{\sqrt{2}}$, vary the frequency again until the output voltage equals $V_{cutoff}$. This frequency is the cut off frequency. There should be 2 cutoff frequencies on either side of resonant frequency.

vi) Calculate the bandwidth by subtracting the 2 cut off frequencies.

vii) calculate the Q factor

viii) Repeat steps 1-2 for circuit in Fig 2.

CALCULATION :

$L = 100\,mH = 0.1\,H$

$C = 0.25\,\mu F = 0.25 \times 10^{-6}\,F$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(0.25\times10^{-6})}}$$

$$f_0 = \frac{1}{2\pi\sqrt{1.58\times10^{-4}}}$$

$$f_0 = \frac{1}{9.92\times10^{-4}}$$

$$f_0 = 1000\ HZ$$

Maximum current (series) and minimum current (parallel) occur at 1000 HZ.

$$BW = \frac{R}{2\pi L} = \frac{330}{2\times3.14\times0.1} = \frac{330}{0.628} = 525.4$$

$$\cancel{BW = f_2 - f_1 =}$$

$$Q = \frac{f_0}{BW} = \frac{1000}{525.4} = 1.903$$

MARKS SPLIT-UP :

| S. NO | DESCRIPTION | MARKS ALLOTED | MARKS OBTAINED |
|---|---|---|---|
| 1 | Experimental setup / connections | 20 | 20 |
| 2 | observation / Readings | 20 | 20 |
| 3 | calculations / Results | 20 | 20 |
| 4 | viva | 20 | 20 |
| 5 | Record | 20 | 20 |
| | | Total | 100 |

RESULT :

Thus the phenomenon of resonance in RLC circuit was studied & the following were determined using Multisim simulator.

Series RLC circuit :

Resonant frequency = 1000

Bandwidth = 525.4

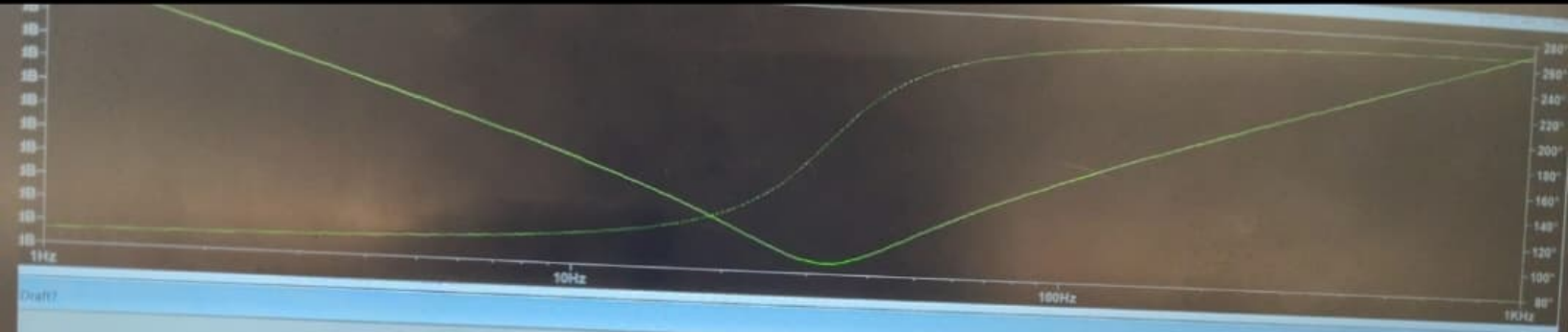
Q factor = 1.903

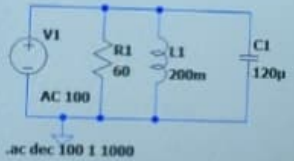
Parallel RLC circuit :

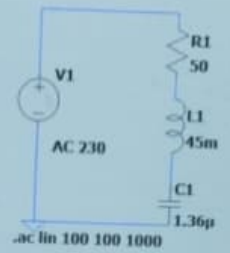
Resonant frequency = 1000

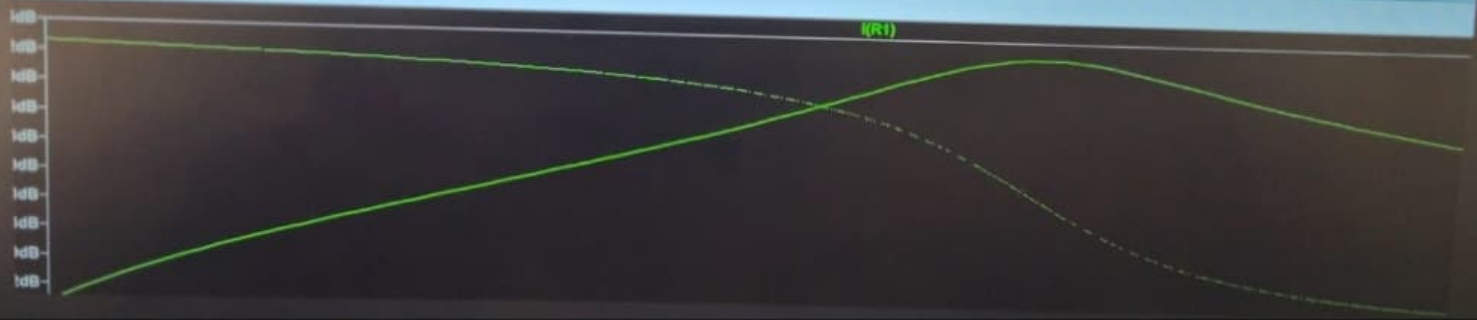
Band width = 525.4

Q factor = 1.903


1Hz
10Hz
100Hz
1KHz


V1
AC 100
R1
60
L1
200m
C1
120µ
.ac dec 100 1 1000


R1
50
V1
L1
45m
AC 230
C1
1.36µ
.ac lin 100 100 1000


I(R1)
1dB
1dB
1dB
1dB
1dB
1dB
1dB
1dB
1dB
1dB